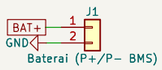
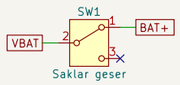
1. DAYA (wiring diagram blok 1)

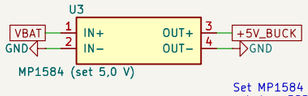
J1 = P+/P− BMS (baterai 2S di rangka)





SW1: kaki tengah (B) = VBAT

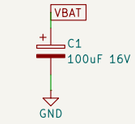
D1: cegah arus balik saat USB dicolok



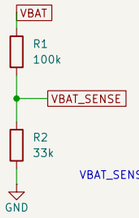


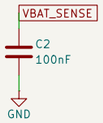
Set MP1584 ke 5,0 V dulu sebelum ESP32 dipasang

PWR_FLAG: daya VBAT, +5V, GND datang dari konektor/dioda

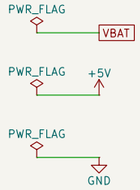


C1 dekat pin VM TB6612

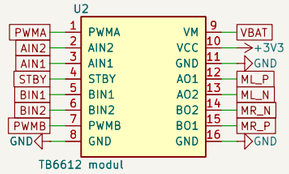


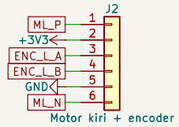


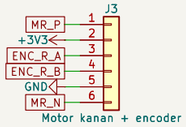
VBAT_SENSE ke GPIO39 (ADC1)



2. DRIVER MOTOR + MOTOR (wiring diagram blok 2)

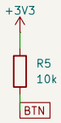




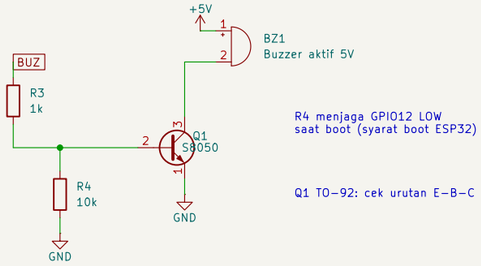


3. BOARD INTERFACE (OLED + 5 TOMBOL), BUZZER

J6: board interface ditumpuk di atas (OLED + tangga resistor 5 tombol)



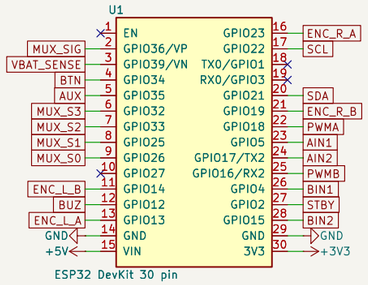
R5 wajib: GPIO34 tanpa pull-up internal



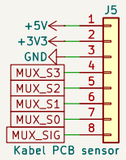
R4 menjaga GPIO12 LOW saat boot (syarat boot ESP32)

Q1 TO-92: cek urutan E-B-C

4. ESP32 DEVKIT (otak)



5. KABEL KE PCB SENSOR (wiring diagram blok 4)



Dibuat oleh hardware/tools/gen_sch_utama.py, pin ESP32 sesuai CLAUDE.md
Panduan per blok: docs/wiring-diagram.html
P01 Kel.5 PjBL DTEO FV ITS

Sheet: /
File: pcb-utama.kicad_sch

Title: LTM — PCB utama

Size: A3	Date: 2026-11-03	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1